JASON CHIGABATIA

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EDUCATION

Grinnell College Grinnell, IA
B.A. in Computer Science, concentration in Linguistics Expected May 2028
Coursework: Data Structures & Algorithms, Analysis of Algorithms, Operating Systems & Parallel Algorithms, Computer Organization & Architecture, Discrete Structures, Automata, Formal Languages, and Computational Complexity

EXPERIENCE

Mid City Engineering Chicago, IL
Software Engineer Intern May 2026 – Aug 2026

- Built and deployed **two internal dashboards** (sales and manufacturing) in Next.js on Odoo 19's JSON-RPC API, surfacing live KPIs such as late orders, missed deadlines, and parts availability; used by the CEO, ~10 engineers, and ~5 sales staff.
- Added server-side in-memory **TTL caching** across 8 Odoo JSON-RPC endpoints, cutting redundant ERP calls from continuously polling dashboards and improving response times.
- Built a Node.js and Bash CLI that deploys and manages **3 Next.js apps** on a live Ubuntu server, automating port allocation, PM2 process management, Nginx configuration, and rollback on failed builds.
- Built the embedded display stack for an ESP32-C3 in ESP-IDF, integrating the SPI bus, GC9A01 round LCD, LVGL, and a SPIFFS filesystem, to deliver a working boot-animation prototype for a **commercial radar detector product line**.
- Fit frame decoding into **~150KB of SRAM** on an MCU without PSRAM by allocating memory per frame.
- Set up Odoo IoT ZPL label printing to Zebra printers over CUPS, reimaging Raspberry Pi devices and migrating a live IoT box.

P3+Uplift Des Moines, IA
Software Engineer Intern Jun 2025 – Aug 2025

- Trained a **TensorFlow CNN** on ~300 labeled images to detect livestock feed-flow anomalies, targeting an estimated **$150K in annual losses** from manual feed-level checks across client farms.
- Built a prototype AWS Lambda alerting function designed to process **~50,000 daily sensor readings** at 5-minute intervals.
- Monitored CloudWatch logs to catch recursive Lambda triggers before they became **runaway invocation loops**, and wrote tests to validate pipeline behavior.
- Presented the prototype and architecture to **board members** at the consulting firm and client company, tying model results to the ~$150K in annual losses so non-engineers could evaluate the business case.

PROJECTS

Watch My Tokens | *ShellHacks 2026* | GitHub *C, ESP-IDF, LVGL, Node.js, Express*

- Built a monitor-mounted **ESP32-C3 pager for Claude Code**: a PreToolUse hook sends each shell command to a Node.js bridge that reads it aloud and relays it over USB serial to a round touchscreen, where a tap approves and a hold denies.
- Wrote the **ESP-IDF firmware** (LVGL 9, GC9A01 display over SPI, CST816D touch over I²C) and a line-based serial protocol; debugged an undocumented pinout, BGR vs. RGB color inversion, and a mirrored display.
- Made every failure path (25 s timeout, bridge offline, concurrent requests) fall back to Claude Code's normal permission prompt instead of auto-approving.

Fissure | *Hack Dearborn 2026* | GitHub *Python, scikit-learn, Next.js, IBM watsonx*

- Built a tool that spots vehicle defects early by grouping NHTSA owner complaints by what drivers wrote; in backtests it flagged a Ford steering defect **3 years before the recall**.
- Built an **unsupervised NLP pipeline** in Python (TF-IDF, KMeans) and validated it against six real vehicle recalls.
- Shipped a Next.js app where **IBM Granite** matches a driver's description to a known defect and rejects unrelated ones.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL, Bash
Frameworks & Libraries: Node.js, Express, Next.js, React, TensorFlow, scikit-learn, pandas, LVGL
Infrastructure & Tools: Linux, Nginx, PM2, SSH, Git, Vercel, AWS Lambda, AWS CloudWatch, IBM watsonx, Vitest
Embedded & Systems: ESP-IDF, ESP32-C3, FreeRTOS, SPI, SPIFFS
Databases: PostgreSQL, Supabase

LEADERSHIP & ACTIVITIES

Freesound Student Organization | *Team Lead & Manager* Aug 2024 – Present

- Co-manage a **~$50K semesterly budget**, equipment access, and live sound mixing on a 7-person leadership team serving 300–400 members across 4+ major events per semester.

ColorStack | *Outreach Marketing Chair* Dec 2024 – Present

- Plan weekly tech talks and host recruiter sessions and alumni panels for underrepresented students in computing.